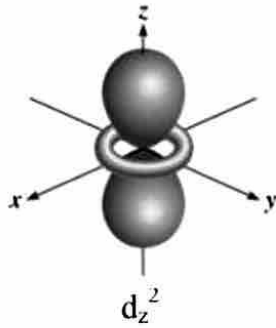
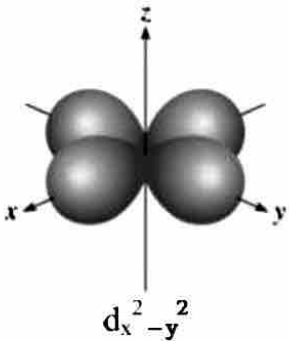
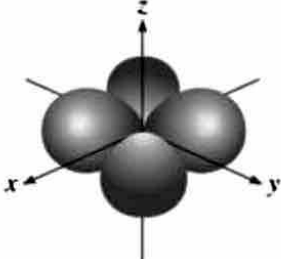
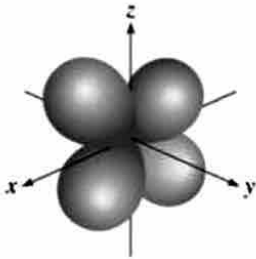
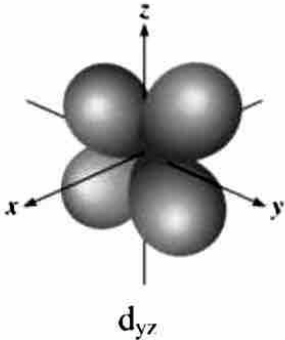
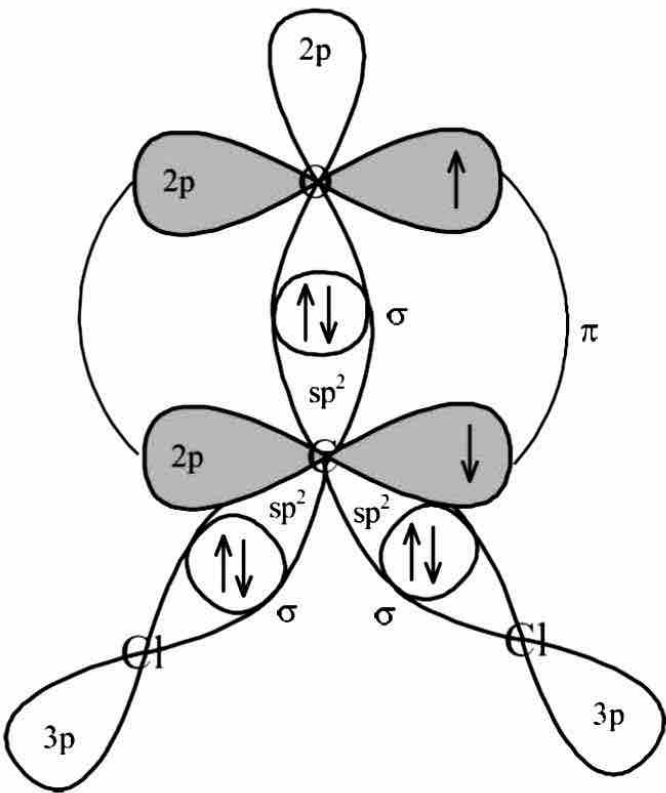


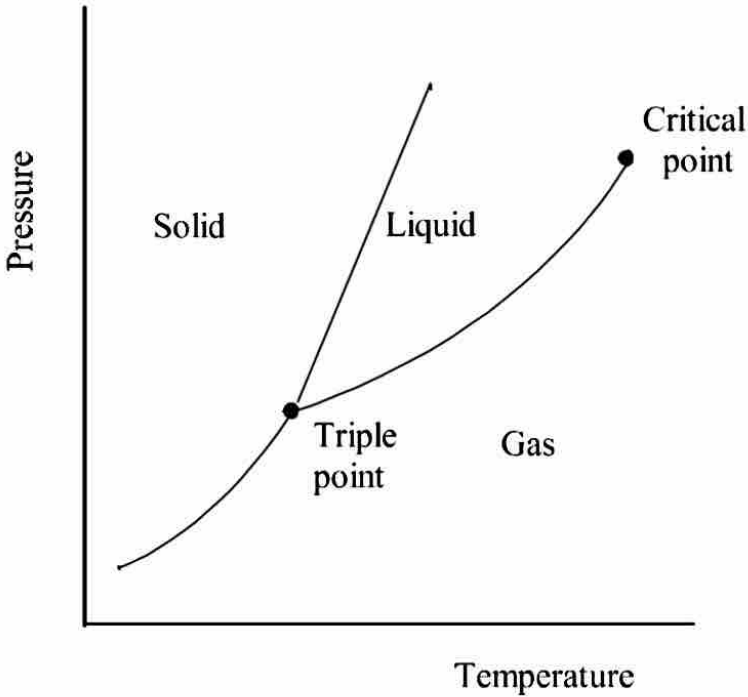
NO.	PART	SCHEME																		
1	(a)(i)	$^{28}_{14}\text{Si}$																		
	(a)(ii)	$\begin{aligned}\text{Number of Si atom} &= \frac{50}{122.1} \times 6.02 \times 10^{23} \\ &= 2.47 \times 10^{23}\end{aligned}$																		
	(b)	$\begin{aligned}\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- &\rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O} \\ 5\text{Fe}^{2+} &\rightarrow 5\text{Fe}^{3+} + 5\text{e}^- \\ \hline \text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} &\rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}\end{aligned}$																		
	(c)	<table><tr><td></td><td>Potassium</td><td>Chromium</td><td>Oxygen</td></tr><tr><td>Mass(g)</td><td>26.58</td><td>35.35</td><td>38.07</td></tr><tr><td>Mol</td><td>$\frac{26.58}{39.1} = 0.6798$</td><td>$\frac{35.35}{52.0} = 0.6798$</td><td>$\frac{38.07}{16.0} = 2.3794$</td></tr><tr><td rowspan="2">Simplest ratio</td><td>$\frac{0.6798}{0.6798} = 1$</td><td>$\frac{0.6798}{0.6798} = 1$</td><td>$\frac{2.3794}{0.6798} = 3.5$</td></tr><tr><td>$1 \times 2 = 2$</td><td>$1 \times 2 = 2$</td><td>$3.5 \times 2 = 7$</td></tr></table> Empirical formula = K₂Cr₂O₇ $\begin{aligned}n &= \frac{\text{Molar mass}}{\text{Empirical mass}} \\ &= \frac{294.20}{294.2} \\ &= \mathbf{1}\end{aligned}$ Molecular formula = K₂Cr₂O₇ Compound X = Potassium dichromate		Potassium	Chromium	Oxygen	Mass(g)	26.58	35.35	38.07	Mol	$\frac{26.58}{39.1} = 0.6798$	$\frac{35.35}{52.0} = 0.6798$	$\frac{38.07}{16.0} = 2.3794$	Simplest ratio	$\frac{0.6798}{0.6798} = 1$	$\frac{0.6798}{0.6798} = 1$	$\frac{2.3794}{0.6798} = 3.5$	$1 \times 2 = 2$	$1 \times 2 = 2$
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	$1 \times 2 = 2$	$1 \times 2 = 2$	$3.5 \times 2 = 7$																	

NO.	PART	SCHEME
	(d)(i)	$\text{Mole of NO}_2 = \frac{100.0 \text{ g}}{46.0 \text{ g mol}^{-1}} \quad @ \quad 2.174 \text{ mol}$ $\text{Mole of H}_2\text{O} = \frac{50.0 \text{ g}}{18.0 \text{ g mol}^{-1}} \quad @ \quad 2.778 \text{ mol}$ $3\text{NO}_2 + \text{H}_2\text{O} \rightarrow 2\text{HNO}_3 + \text{NO} \quad @$ $3 \text{ mol NO}_2 \equiv 1 \text{ mol H}_2\text{O}$ $2.174 \text{ mol NO}_2 \equiv \frac{2.174}{3} \quad 1 \quad \text{H}_2\text{O}$ $= 0.725 \text{ mol H}_2\text{O}$ Since mole H₂O needed is smaller than mole H₂O present, $\therefore$ NO₂ is a limiting reactant
	(d)(ii)	From the reaction: $3 \text{ mol NO}_2 \equiv 2 \text{ mol HNO}_3 \quad @$ $2.174 \text{ mol NO}_2 \equiv \frac{2.174}{3} \quad \text{HNO}_3$ $= 1.449 \text{ mol HNO}_3$ Mass of HNO₃ produced = $1.449 \text{ mol} \times 63.0 \text{ g mol}^{-1} \quad @$ $= 91.3 \text{ g}$ $\therefore \text{Percentage yield} = \frac{80.0 \text{ g}}{91.3 \text{ g}} \times 100\%$ $= 87.6 \%$
		TOTAL
		MAXIMUM

NO.	PART	SCHEME
	(b)	The electron is in 3d orbital because $\ell = 2$  Correct shape and complete label (axis and types): 5 correct = 3M 3 @ 4 correct = 2M 2 correct = 1M
		TOTAL

NO.	PART	SCHEME
3	(a)	The three possible structures: $\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \\ \text{:}\ddot{\text{Cl}}_1=\text{C}-\ddot{\text{Cl}}_2\text{:} \end{array}$ I $\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \\ \text{:}\ddot{\text{Cl}}_1-\text{C}-\ddot{\text{Cl}}_2\text{:} \end{array}$ II $\begin{array}{c} \text{:}\ddot{\text{O}}\text{:} \\ \\ \text{:}\ddot{\text{Cl}}_1-\text{C}=\ddot{\text{Cl}}_2\text{:} \end{array}$ III
	(b)	Formal Charge = no. of valence e^- - no. of lone pair e^- - $\frac{1}{2}$ (no. of bonding e^-) Structure I: $\text{Cl}_1 = +1, \text{Cl}_2 = 0, \text{C} = 0, \text{O} = -1$ Structure II: $\text{Cl}_1 = 0, \text{Cl}_2 = 0, \text{C} = 0, \text{O} = 0$ Structure III: $\text{Cl}_1 = 0, \text{Cl}_2 = +1, \text{C} = 0, \text{O} = -1$ The most plausible structure = Structure II. Reason: All atoms have formal charge equal to zero @ lowest formal charge
	(c)	Hybrid orbital of C in structure II = sp^2 Ground state: $\begin{array}{ c } \hline \uparrow\downarrow \\ \hline \end{array} \quad \begin{array}{ c c c } \hline \uparrow & \uparrow & \\ \hline \end{array}$ $\quad \quad \quad 2s \quad \quad \quad 2p$ Excited state: $\begin{array}{ c } \hline \uparrow \\ \hline \end{array} \quad \begin{array}{ c c c } \hline \uparrow & \uparrow & \uparrow \\ \hline \end{array}$ $\quad \quad \quad 2s \quad \quad \quad 2p$ Hybrid state: $\begin{array}{ c c c } \hline \uparrow & \uparrow & \uparrow \\ \hline \end{array} \quad \begin{array}{ c } \hline \uparrow \\ \hline \end{array}$ $\quad \quad \quad sp^2 \quad \quad \quad 2p$

NO.	PART	SCHEME
	(d)	Molecular shape of COCl_2 = Trigonal planar  Shape of sp^2 for C, p for O and Cl Label of sp^2 and p 3 overlapping of sp^2-p orbital and label σ bond 1 overlapping of p-p orbital and label π bond
		TOTAL

NO.	PART	SCHEME
4	(a)	 Triple point Critical point Shape and slope Complete label (phase and axis)
	(b)	$n_{\text{He}} = 2/4$ @ 0.5 $n_{\text{O}_2} = 61/32$ @ 1.91 $P = \frac{nRT}{V}$ @ substitution value $P_{\text{He}} = \frac{0.5 \times 0.08206 \times 298}{5}$ $= \mathbf{2.45 \text{ atm}}$ (insist unit) $P_{\text{O}_2} = \frac{1.91 \times 0.08206 \times 298}{5}$ $= \mathbf{9.32 \text{ atm}}$ (insist unit) $P_{\text{Total}} = 2.45 + 9.32$ $= \mathbf{11.77 \text{ atm}}$ (insist unit)

NO.	PART	SCHEME
		<u>ALTERNATIVE ANSWER:</u> $ \begin{aligned} n_T &= n_{\text{He}} + n_{\text{O}_2} \quad @ \\ &= 0.5 + 1.91 \\ &= 2.41 \text{ mol} \end{aligned} $ $ \begin{aligned} P_T &= \frac{n_T RT}{V} \quad @ \\ &= \frac{2.41 \times 0.08206 \times 298}{5} \\ &= 11.77 \text{ atm (insist unit)} \end{aligned} $ $ P_{\text{He}} = \frac{n_{\text{He}}}{n_T} \times P_T = \frac{0.5}{1} \times 11.77 = $ $ P_{\text{O}_2} = 11.77 - 2.45 = 9.32 \text{ atm (insist unit)} $
		TOTAL
		MAXIMUM

NO.	PART	SCHEME
5	(a)	Reduce the reaction temperature Because the forward reaction is exothermic @ Increase the reaction pressure Because the number of mole of product is smaller than that of reactant @ Continuous removal of NH₃ Because concentration of NH₃ decrease thus equilibrium will shift to favour the production of NH₃ @ Addition of H₂ or N₂ Because the concentration of H₂ or N₂ increase thus equilibrium will shift to favour the production of NH₃ *Any TWO changes
	(b)	$Q_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]}$ $= \frac{(0.15)^2}{(0.2)^2 (8.55 \times 10^{-3})}$ $= 65.8$ Since $Q_c > K_c$, the reaction favours the reverse reaction @ net reaction occurs to the left
		TOTAL

NO.	PART	SCHEME																				
6	(a)	<table><tr><td></td><td>CH₃COOH</td><td>⇌</td><td>H⁺</td><td>CH₃COO⁻</td></tr><tr><td>Initial []</td><td>0.5</td><td></td><td>0</td><td>0</td></tr><tr><td>Change []</td><td>-x</td><td></td><td>+x</td><td>+x</td></tr><tr><td>Equilibrium []</td><td>0.5 - x</td><td></td><td>x</td><td>x</td></tr></table>		CH ₃ COOH	⇌	H ⁺	CH ₃ COO ⁻	Initial []	0.5		0	0	Change []	-x		+x	+x	Equilibrium []	0.5 - x		x	x
			CH ₃ COOH	⇌	H ⁺	CH ₃ COO ⁻																
		Initial []	0.5		0	0																
		Change []	-x		+x	+x																
		Equilibrium []	0.5 - x		x	x																
$K_a = \frac{[H^+][CH_3COO^-]}{[CH_3COOH]} \quad @$																						
$1.8 \times 10^{-5} = \frac{x^2}{(0.5 - x)}$																						
Since x is very small, x is neglected @ quadratic equation																						
		$\frac{x^2}{(0.5)} = 1.8 \times 10^{-5}$																				
		$x = 3.0 \times 10^{-3} = [H^+]$																				
		pH = -log [H ⁺] @																				
		= -log (3 x 10 ⁻³)																				
		= 2.52																				
	(b)(i)	CH ₃ COOH + NaOH → CH ₃ COONa + H ₂ O 1 mole of CH ₃ COOH produced 1 mole of CH ₃ COONa Mole of CH ₃ COOH = 0.5 × 0.025 = 1.25 × 10 ⁻² mol Therefore the number of mol of CH ₃ COONa = 1.25 × 10 ⁻² mol The concentration of CH ₃ COONa solution: $= \frac{1.25 \times 10^{-2}}{0.05}$ $= 0.25 \text{ M}$																				

NO.	PART	SCHEME
	(b)(ii)	$\text{CH}_3\text{COONa} \rightarrow \text{CH}_3\text{COO}^- + \text{Na}^+$ Hydrolysis of CH_3COO^- $\text{CH}_3\text{COO}^- + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COOH} + \text{OH}^-$ Since OH is present $\therefore \text{pH} > 7$
		TOTAL
		MAXIMUM